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MIL-R-29530(EC)
20 October 1988

MILITARY SPECIFICATION

RECORDER-REPRODUCER, DIGITAL RD-433( )/SSH

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, test, manufacture, and acceptance requirements for the Recorder-Reproducer, Digital, RD-433( )/SSH, critical item, hereinafter referred to as RD-433.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment And Systems, Requirements For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command, SPAWAR PMW152, Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 7025

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MIL-E-16400G	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification For
MIL-P-28847	Processing Sets, Data, AN/UYK-20(V) And AN/UYK-20X(V)
MIL-M-38510	Microcircuit, General Specification For

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

ELEX-R-399	Recorder-Reproducer, Digital, Magnetic Disk, RD-433/SSH

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation

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MIL-STD-781	Reliability Testing For Engineering Development, Qualification, And Production
MIL-STD-810C	Environmental Test Methods
MIL-STD-883	Test Methods And Procedures For Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
DoD-STD-1686	Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
DoD-STD-2000-1	Soldering Technology, High Quality/High Reliability
DoD-STD-2000-2	Part And Component Mounting For High Quality/High Reliability Soldered Electrical And Electronic Assemblies
DoD-STD-2000-3	Criteria For High Quality/High Reliability Soldering Technology

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NATIONAL SECURITY AGENCY

NACSIM 5100A	Compromising Emanations Test Standards Electromagnetic
NACSEM 5201	Compromising Emanations Design Handbook

ASSISTANT SECRETARY OF THE NAVY (SHIPBUILDING AND LOGISTICS)

NAVMAT P 4855.1	Navy Power Supply Reliability, Design And Manufacturing Guidelines

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SPAWAR

EE640-AA-OMI-01A/E106	Technical Manual, Digital Magnetic Disc Recorder-Reproducer RD-433-SSH

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DRAWINGS

SINGER-LIBRASCOPE

FAT-L-192-000-478	Mass Memory System Model RD-433( )/SSH Factory Acceptance Test
L510-000-032	Battery Assembly
L535-301-831	Circuit Card Assembly, Power Supply, Battery Charger
L535-301-828	Circuit Assembly, DRAM, Mega B.F.

(Copies of drawings required by contractors in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

PUBLICATION

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels, And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The RD-433 shall be in accordance with MIL-E-16400G to the extent specified herein. The RD-433 shall form a part of various communication subsystems and shall be used in conjunction with the Computer, AN/UYK-20, in accordance with MIL-P-28847, to provide additional high speed random access memory for the storage and retrieval of program modules, system data base, and message traffic. The RD-433 shall be a solid-state memory replacement for the RD-433 rotating disk memory, as produced in accordance with ELEX-R-399, and shall be installed interchangeably with the RD-433 at the system level.

3.1.1 Major physical components. The RD-433 shall consist of two major components, a memory unit and a controller. The memory unit shall consist of a bank of integrated circuit (IC) memory and associated timing and serialization circuitry. The controller shall contain the necessary power supplies, interface circuitry, input and output (I/O) buffers, and all status and control logic required between the Computer, AN/UYK-20 and the RD-433 memory unit.

3.1.2 System software interface. The RD-433 shall operate in conjunction with all existing depot level maintenance software without modification, as implemented on the AN/UYK-20. The RD-433 shall provide two modes of operation: a slow mode, which simulates the rotating memory of the RD-433 and the average access time for a sector shall not exceed 20 milliseconds (ms), and a fast mode with not more than 2.1 ms access time.

3.1.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Characteristics. Characteristics shall be as specified in 3.2.1 through 3.2.2.3.

3.2.1 Performance. Performance characteristics shall be as specified in 3.2.1.1 through 3.2.1.6.

3.2.1.1 Capacity. The RD-433 shall provide a storage capacity of one million 16-bit words, exclusive of formatting, checksum or overhead, to the AN/UYK-20 for storage of programs and data.

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3.2.1.2 Organization. The RD-433 shall be organized into individually accessible sectors of 256 16-bit words each. Sectors may contain additional overhead, and each sector shall contain a cyclic redundancy check (CRC) for the data recorded in that sector. The CRC shall be computed when data are written into each sector and shall be verified when data are read out of a sector.

3.2.1.3 Command interface. The RD-433 shall be completely compatible with the command and data protocol as specified in the Overall function description paragraph of SPAWAR EE640-AA-OMI-01A/E106. The RD-433 controller shall provide buffering for all I/O data transfers to and from the AN/UYK-20 to allow for short term variations in the nominal 125 k words per second transfer rate. Both buffers shall be first-in, first-out buffers, with a minimum of 256 words. Timing errors resulting from buffer underflow or overflow shall be reported as specified in SPAWAR EE640-AA-OMI-01A/E106.

3.2 1.4 Speed. The average access time for a sector shall not exceed 20 ms in the slow mode. In the fast mode, no access delays shall be required, with data access limited only by the data transfer rate. The data transfer rate between the RD-433 and the AN/UYK-20 shall be a nominal 125 k words per second.

3.2.1.5 Write protection. The RD-433 shall provide a means to write protect all sectors below a specified address. The range protected shall be from zero to an upper address specified by manual switches. A separate protect override switch shall be provided. The RD-433 shall not allow write operations to the protected area unless the protect override switch is enabled.

3.2.1.6 Memory backup. Battery backup shall be provided in the memory subassembly, providing a minimum of 4 hours of data integrity from loss of alternating current (AC) input power.

3.2.2 Physical requirements. Physical requirements shall be as specified in 3.2.2.1 through 3.2.2.3.

3.2.2.1 Electrical interface. The RD-433 interface to the AN/UYK-20 shall be a 16-bit parallel Type B, Category I, interface as specified in MIL-STD-1397.

3.2.2.2 Physical interface. The physical interface between the RD-433 and the AN/UYK-20 shall be as specified in the general requirements of MIL-STD-1397. The two I/O connectors on the RD-433 shall be modified Type 436-0004-105 Malco, Incorporated connectors, or equivalent, which conform to Type 5593563 Univac connectors, or equivalent. The mechanical configuration of the connectors shall be compatible for use with standard AN/UYK-20 parallel I/O cables which use Univac Type 7101943-02 (input) and Univac Type 7101943-03 (output) connectors, or equivalent.

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3.2.2.3 Dimensions and weight. The RD-433 shall conform to dimensions specified in the outline and installation drawing in SPAWAR EE640-AA-OMI-01A/E106, and conform to requirements for mounting in a standard 19-inch rack as specified in EIA RS-310-C-77. Rack slides shall be provided with the RD-433, capable of supporting the RD-433 in the fully extended position. Total weight shall not exceed 200 pounds.

3.3 Environmental conditions. Environmental conditions shall be as specified in 3.3.1 through 3.3.6.

3.3.1 Shock. The RD-433 shall conform to the shock requirements of MIL-S-901, Grade A, Class II, Type A for lightweight equipment. During high impact shock the RD-433 shall not suffer any nonrecoverable loss of stored data and there shall be no loss of I/O data during transfers. Manual intervention shall not be required in the event of momentary malfunction.

3.3.2 Vibration. The RD-433 shall conform to the vibration requirements of MIL-STD-167-1, Type I.

3.3.3 Nonoperating temperature. The RD-433 shall not be damaged nor shall operational performance be degraded by exposure to a temperature range from -62° Celsius (C) to +71°C.

3.3.4 Operating temperature. The RD-433 shall maintain operational performance over a range of 0°C to 50°C. The RD-433 shall be operational, with degradation beyond 55°C when the battle-short switch is enabled.

3.3.5 Humidity. The RD-433 shall maintain operational performance when exposed to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions where condensation occurs in or on the equipment in the form of both water and frost.

3.3.6 Inclination. The RD-433 shall conform to the Inclination paragraph of MIL-E-16400G. Inclination angle shall be 45 degrees.

3.4 Parts, materials, and processes. Parts, materials, and processes shall be as specified in 3.4.1 through 3.4.3.2.

3.4.1 Parts. Parts shall be as specified in 3.4.1.1 through 3.4.1.5.

3.4.1.1 Standard parts. Standard parts shall be in accordance with the Standard parts paragraph of MIL-E-16400G. Procuring activity approval of nonstandard parts shall be obtained before the unit can be accepted.

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3.4.1.2 Interchangeability. The RD-433 subassemblies shall be form, fit, and function interchangeable with the subassemblies of the RD-433, as produced in accordance with ELEX-R-399, at the levels specified in a through d:

a. The RD-433 chassis, including all mounting holes and studs, and front screw assemblies
b. Components specified in SPAWAR EE640-AA-OMI-01A/E106, Parts list chapter, List of major components table
c. Wiring harnesses between major components specified
d. All circuit card assemblies in major component A1A2, Electronic Assembly, as specified in SPAWAR EE640-AA-OMI-01A/E106, Parts list chapter, Mass memory parts list table. The solid-state memory assembly shall replace the disk memory assembly in form, fit, and function.

3.4.1.2.1 Solid-state memory. The solid-state memory shall be interchangeable with the disk memory for RD-433s produced in accordance with ELEX-R-399. The solid-state memory shall have the same physical appearance, including mounting features, as the disk memory, with the exception of the Battery Charge and Power/Charging indicators. The impact and humidity indicators provided with the disk memory are not required for the solid-state memory.

3.4.1.2.2 Solid-state memory components. A solid-state memory shall replace the disk memory assembly, as a unit, in form, fit, and function. Memory subcomponents, produced in accordance with SINGER-LIBRASCOPE Drawings (DWGS) L510-000-032, L535-301-831, and L535-301-828, shall consist of a battery assembly, a power supply and battery charge, printed circuit board (PCB), and a 2-megabyte integrated circuit memory PCB.

3.4.1.3 Electrostatic discharge (ESD). The ESD control program shall be in accordance with DoD-STD-1686.

3.4.1.4 Microelectronic devices. All microelectronic devices employed in the RD-433 shall be qualified in accordance with Class B requirements of MIL-M-38510 or tested in accordance with MIL-STD-883, Class B. Government approval will be required for use of devices utilizing current technology, but not covered by MIL-M-38510. Request for approval of these devices shall be in accordance with MIL-STD-965, Procedure I.

3.4.2 Unacceptable materials. Equipment design shall not include polychlorinated biphenyls, asbestos and asbestos compounds, fragile or brittle materials, beryllium and beryllium compounds unless so identified, and lithium and lithium compounds unless specifically approved by the procuring activity. The Unacceptable materials paragraph of MIL-E-16400G shall apply.

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3.4.3 Processes. Processes shall be as specified in 3.4.3.1 and 3.4.3.2.

3.4.3.1 Protective coating. Unless contained in hermetically sealed units, parts shall be resistant to corrosion in accordance with MIL-STD-454, Requirement 15. The equipment interior and exterior surfaces shall be painted in accordance with the Painting paragraph of MIL-E-16400G.

3.4.3.2 Soldering. Soldering shall be in accordance with DoD-STD-2000-1, DoD-STD-2000-2, and DoD-STD-2000-3.

3.5 Design and construction. Design and construction shall be as specified in 3.5.1 through 3.5.2.2.

3.5.1 Electrical design. Electrical design shall be as specified in 3.5.1.1 through 3.5.1.2.14.

3.5.1.1 Power. The RD-433 shall conform to the Power source variations and Power factor paragraphs of MIL-E-16400G.

3.5.1.2 Power supply. The power supply shall be as specified in 3.5.1.2.1 through 3.5.1.2.14.

3.5.1.2.1 Power density. Power density shall be output power divided by power supply envelope volume, including electromagnetic interference (EMI) filtering where required. Power density exceeding 2 watts per cubic inch ($W/in.^3$) for switching mode power supplies or 0.5 $W/in.^3$ for linear power supplies shall require the approval of the procuring activity.

3.5.1.2.2 Junction temperature. Junction temperature of semiconductor and microelectronic devices (diodes, transistors, hybrid, IC, and so forth), shall not exceed +110°C, under worst-case conditions.

3.4.1.5 PCBs. PCBs shall conform to MIL-STD-454, Requirement 17 and MIL-STD-275.

3.5.1.2.3 Hot spot. Case or hot spot temperatures shall not exceed 40°C rise above ambient temperature with a maximum temperature of +110°C for parts less than or equal to 3 watts of dissipation; 55°C rise above ambient with a maximum temperature of +125°C for parts greater than 3 watts of dissipation; 30°C rise above part ambient with maximum temperature of +100°C for transformers; and 10°C rise above ambient, due to self-heating, with a maximum temperature of 85°C for capacitors.

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3.5.1.2.4 Nonstandard parts. The use of nonstandard parts shall not be prohibited, provided there are at least two independent sources. Procuring activity approval of the nonstandard parts shall be obtained before the unit can be accepted.

3.5.1.2.5 Maintenance. The power supplies shall:

a. Be designed to be maintainable at either intermediate or depot level

b. Not be limited by design to repair by the manufacturer

3.5.1.2.6 Encapsulation. The power supply shall not be encapsulated or embedded (potted) unless it can be shown by analysis and test to be necessary for heat removal or dissipation. This requirement shall not exclude conformal coating.

3.5.1.2.7 Manufacturing. Power supply manufacturing shall include random vibration and temperature cycling of every unit, under full electrical load; this may be done in the end item but shall be performed on a unit basis for spares.

3.5.1.2.8 Cooling. Conduction cooling to a heat exchanger shall be the preferred method for heat removal. Static convection cooling should be avoided whenever practical.

3.5.1.2.9 Design. The power supply shall be designed to operate from Type I power specified in DoD-STD-1399, Section 300.

3.5.1.2.10 EMI. EMI power supply requirements shall be tailored to enable the RD-433 to conform to the requirements of 3.6. EMI requirements shall permit the purchase of the power supply separately from the RD-433.

3.5.1.2.11 Mean-time-between-failures (MTBF). The MTBF shall be not less than 40,000 hours for Naval sheltered equipment at 55 C.

3.5.1.2.12 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.5.1.2.13 Overload. The power supply shall be designed with a reserve load capacity of 20 percent.

3.5.1.2.14 Design guidance. Additional guidance for design and manufacturing of power supplies is contained in NAVMAT P 4855-1.

3.5.2 Mechanical design. Mechanical design shall be as specified in 3.5.2.1 and 3.5.2.2.

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3.5.2.1 Enclosure. The RD-433 shall conform to the dripproof 45 degrees requirements specified in MIL-STD-108. The enclosure shall protect personnel from electrical and physical injury during normal operation and required maintenance activities. Connectors shall be mounted in the back of the enclosure. All connectors shall conform to the requirements of MIL-E-16400G.

3.5.2.2 Controls and indicators. All controls and indicators shall be positioned as specified in SPAWAR EE640-OMI-01A/E106, the Controls and indicator paragraph of the Operations chapter, with the exception of indicators mounted on the disk memory assembly. Controls and indicators shall conform to the Controls, indicators and panel layout paragraph of MIL-E-16400G, and be of color and style equivalent to that specified in SPAWAR EE640-AA-OMI-01A/E106. The abbreviations of MIL-STD-12 shall be used for equipment labeling where applicable.

3.6 EMI. The RD-433 shall be self-compatible and operate compatibly in its intended electromagnetic environment. As a minimum, the RD-433 shall conform to the requirements of MIL-STD-461.

3.7 TEMPEST. The RD-433 shall incorporate TEMPEST control concepts of NACSEM 5201 to control the radiation of compromising energy from the unit. The RD-433 shall operate with black power as specified in NACSIM 5100A.

3.8 Identification and marking. Nameplates, markings, and labels shall be in accordance with the Identification and marking paragraph of MIL-E-16400G. Each replaceable module shall be marked as specified in a through c:

a. Item nomenclature
b. Serial number
c. National stock number or manufacturer's part number

An identification plate, in accordance with MIL-STD-130, shall be installed on the front of each unit. The plate shall include, as a minimum, the manufacturer's name, model designation, contract number, and serial number and indicate that the equipment was manufactured for the Space and Naval Warfare Systems Command.

3.9 Safety. The design of the RD-433 shall conform to the Safety criteria paragraph of MIL-E-16400G, except as specified in 3.9.1 through 3.9.6

3.9.1 Voltage protection. With the RD-433 assembled and in operation, personnel shall be protected from contact with potentials in excess of 30 volts (V) to ground, chassis or frame, including potentials on charged capacitors. Incoming power line connections shall be given physical protection against accidental contact. Internal protective measures shall be as specified in MIL-STD-454, Requirement 1, the Suitable Internal Protective Measures table.

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3.9.2 Ground potential. The design of the RD-433 shall ensure that all external parts, surfaces and shields are at ground potential at all times during normal operation and maintenance in accordance with MIL-STD-454, Requirement 1. Provisions shall be included for grounding the enclosure to a common ground point within the ship's structure by a suitable ground strap. All grounds within a section or subassembly shall be connected to ground on the enclosure. Circuit grounding shall be in accordance with MIL-F-18870.

3.9.3 Warning labels and markings. All contacts, terminals or like devices, which have potentials in excess of levels specified in MIL-STD-454, Requirement 1, the Suitable Internal Protective Measures table, shall be appropriately marked.

3.9.4 Leakage current equipment. Leakage current shall comply with the Leakage current paragraph of MIL-E-16400G. Leakage current shall not exceed 5 milliamperes for 115 volts alternating current (VAC), 60-hertz (Hz) power.

3.9.5 115-VAC equipment power connections. Equipment power connectors for 115-VAC input power and interconnecting AC power cable connectors A, B, and C. Pin assignment and conductor color code within the equipment shall be identical to RD-433s produced in accordance with ELEX-R-399.

3.9.6 Electrical cable protection. Equipment design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of equipment.

3.10 Reliability. The quantitative parameter for the probability that the RD-433 will perform its intended function for a specified interval shall be expressed as MTBF expressed in continuous hours of operation. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be equal to, or greater than 4400 hours.

3.11 Maintainability. The mean-time-to-repair (MTTR) for the RD-433 shall not exceed 20 minutes. The maximum-corrective-maintenance-time (M_{maxct}) shall not exceed 120 minutes at the 95th percentile when repair is accomplished by replacement of lowest subassemblies (modules, PCBs), or chassis-mounted components (including electronic, electromechanical, and mechanical components). The specified MTTR shall include localization, isolation, disassembly, interchange, reassembly, and checkout of all maintenance tasks.

3.12 Workmanship. Workmanship shall be as specified in 3.12.1 and 3.12.2.

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3.12.1 General workmanship. The equipment, including subassemblies and assemblies shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of MIL-E-16400G. Each solder connection and its associated wiring or leads shall be visually examined.

3.12.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen and temperature cycling screen as specified in 4.10.2.1 and 4.10.2.2. Vibration shall be performed prior to temperature cycling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be defined in MIL-STD-109.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
   1. Production inspection (Group A) (see 4.5.1)
   2. Production control inspection (Group B) (see 4.5.2)
   3. Environmental inspection (Group C) (see 4.5.3)
c. Safety verification (see 4.6.7)
d. Inspection of packaging (see 4.11)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.4 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I. The first article unit shall also be tested by a Government facility to ensure compliance is achieved.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.5.1 through 4.5.4.

4.5.1 Production inspection (Group A). Production inspection shall be conducted on each RD-433 offered for delivery. Production inspection shall comprise examinations and tests which will prove the workmanship and reveal omissions and errors of the production process. The tests shall include functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in Group A of TABLE I.

4.5.2 Production control inspection (Group B). Production inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation, tests which will detect any deterioration of the design by wear of such items as dies, models, jigs, and by substitution of different parts, tests which detect deviations in the processing of materials, tests to determine temperature rise produced in operation and the ability of equipment to withstand such temperature, tests of efficiency, and tests of performance with other equipments. Production control inspection shall be performed on inspection lots that have passed productioninspection (see 4.5.1). Lot acceptance criteria shall be based on a single sampling plan with a zero acceptance number. Production control inspection shall include the examinations and tests specified in Group B of TABLE I.

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TABLE I. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.2.2.3	4.6.1	X			
Weight	3.2.2.3	4.6.1	X	X		
Dimensions						
Parts, materials, and processes	3.4	4.6.1	X	X		
Finish	3.4.3.1	4.6.1	X	X		
Identification and marking	3.8	4.6.1	X	X		
General workmanship	3.12.1	4.6.1, 4.10.1	X	X		
Workmanship screen	3.12.2	4.10.2	X	X		X
Performance	3.2.1	4.6.2	X	X	X	
Preperformance	-	4.6.2	X			
Transient voltage and frequency	3.5.1.1	4.6.4	X		X	
Spike voltage	3.5.1.1	4.6.4	X		X	
Power interruption	3.5.1.1	4.6.4	X		X	
Power and power factor	3.5.1.1	4.6.4	X		X	
Enclosure	3.5.2.1	4.6.2	X			X
Controls and indicators	3.5.2.2	4.6.2	X	X		
Circuits (PCBs)	3.4.1.5	4.6.2	X	X		
Nonoperating temperature	3.3.3	4.6.3.3, 4.6.3.4	X			X
Operating temperature	3.3.4	4.6.3.3, 4.6.3.4	X			X

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TABLE I. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Humidity	3.3.5	4.6.3.5	X			X
Abnormal temperature	3.3.4	4.6.3.4	X		X	
Inclination	3.3.6	4.6.3.6	X			X
Alarm and battle-short	3.3.4, 3.3.5	4.6.3.4	X	X		
Shock	3.3.1	4.6.3.1	X			X
Vibration	3.3.2	4.6.3.2	X			X
Power supply	3.5.1.2	4.6.5	X		X	
EMI	3.6	4.6.6	X			X
TEMPEST	3.7	4.6.6	X			
Safety	3.9	4.6.2, 4.6.7	X	X		
Grounding	3.9.2	4.6.2	X			
Leakage current	3.9.4	4.6.7.1	X		X	
115-VAC power connections	3.9.5	4.6.2	X			
Reliability	3.10	4.8	X			
Maintainability	3.11	4.9	X			

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4.5.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.2.2 Failed units. If a unit fails Group B inspection, the cause of the failure shall be determined. All failures shall be corrected and the test continued.

4.5.3 Environmental inspection (Group C). Environmental inspection shall be performed on a sampling basis as specified in 4.5.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole, life test, simulated service tests, tests of the effects of changes of environment, and tests of the effects of shock, vibration, and inclination.

4.5.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first month's production.

4.5.3.2 Nonconforming environmental sample units. If a unit fails the inspection specified herein, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative of the results thereof and the details of the corrective action taken to correct units which were manufactured under the same conditions, materials, processes, and so forth. The equipment shall then be retested.

4.5.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control or environmental inspections, may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.6 Test methods. Examinations and tests shall be performed in accordance with the test methods of MIL-E-16400G, except as specified in 4.6.1 through 4.6.7.1.2.

4.6.1 Surface examination. The RD-433 shall be examined to determine conformance to the applicable requirements of this specification with respect to finish, markings, and workmanship. In the process of examination, the RD-433 shall not be disassembled in a manner that will affect the performance, durability, or appearance. The examination shall include a check of all operating controls and adjustments.

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4.6.2 Performance test. The RD-433 shall be subjected to a performance test, including all operating controls, to determine conformance to the performance parameters and safety requirements specified herein. The performance test shall include, but shall not be limited to, tests that prove conformance of the RD-433A to the requirements specified herein. The performance test shall consist of those tests specified in SINGER-LIBRASCOPE DWG FAT-L-192-000-478, except where such tests pertain to characteristics unique to the RD-433 rotating memory.

4.6.3 Environmental tests. Environmental tests shall be performed as specified in 4.6.3.1 through 4.6.3.6.

4.6.3.1 Shock. The RD-433 shall be tested to determine conformance to the requirements of 3.3.1. The RD-433 shall be energized and fully operating during and after the shock test. The use of simulated loads such as weights or other dummy masses is not acceptable. Loss of stored data or I/O shall not occur during high impact shock.

4.6.3.2 Vibration. The RD-433 shall be tested to determine conformance to the requirements of 3.3.2. The RD-433 shall be energized and fully operating during and after the vibration test. The use of simulated loads such as weights or dummy masses is not acceptable. Loss of stored data or I/O shall not occur during the test.

4.6.3.3 Low temperature. The low temperature test shall be in accordance with MIL-STD-810C, Method 502.1, except as specified in a through c:

a. Step 2. Storage or nonoperating temperature shall be -62°C +0°C -5°C for a duration of 24 hours.

b. Step 4. Lowest operating temperature shall be 0°C.

c. The combined duration of Step 4 and Step 5 shall be not less than 12 hours, at least 2 hours of which shall be in the stabilized condition prior to conducting the tests.

4.6.3.4 High temperature. The high temperature test shall be in accordance with MIL-STD-810C, Method 501.1, Procedure I, except as specified in a through d:

a. Step 4. Applicable high operating temperature shall be 50°C.

b. The combined duration of Step 4 and Step 5 shall be not less than 12 hours, at least
6 hours of which shall be in the stabilized condition prior to conducting the tests.

c. Step 7. The equipment shall operate for a period of at least 2 hours in the stabilized condition before the Step 7 tests are performed.

d. After the procedures in a through c are performed, the temperature shall be raised to verify operation of the temperature alarm and protection devices at the specified temperatures.

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4.6.3.5 Humidity. The humidity test shall be in accordance with MIL-STD-810C, Method 507.1, Procedure IV, except as specified in a and b:

a. Step 5. Equipment performance measurements shall be made at 50°C.
b. Step 9. The equipment shall be carefully examined in detail to detect evidence of physical degradation. Blistering of organic coatings shall be cause for rejection.

4.6.3.6 Inclination. The RD-433 shall be subjected to the test limits specified in 3.3.6. The RD-433 shall be energized and fully operating during and after the test.

4.6.4 Power. The RD-433 shall be tested to determine conformance to the requirements of 3.5.1.1.

4.6.5 Power supply. The power supply shall be tested with the RD-433, or shall be tested as a unit by duplicating the I/O power of the supply as installed for operation in the RD-433. The tests shall determine conformance to 3.5.1.2.

4.6.6 Integration EMI, electromagnetic compatibility (EMC), and TEMPEST evaluations. One first article unit shall be delivered to the Government facility specified in the contract. The unit shall be used to demonstrate integration of the RD-433 into the overall system. The same unit shall be evaluated for conformance to EMI, EMC, and TEMPEST requirements. The requirements of 3.6 shall be verified, utilizing the test methods of MIL-STD-462. For first article inspection, the Emission and susceptibility requirements for Class A4 equipment and subsystems table of MIL-STD-461 shall be complied with; for Group C inspection, tests CE03 and RE02 shall be performed. Tests CE01, CE03, CS01, CS02, CS03, CS05, CS06, RE01, RE02, RS01, RS02, and RS03 (below-deck limit) are applicable.

4.6.7 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.9.

4.6.7.1 Leakage current test. The leakage current test shall be as specified in 4.6.7.1.1 and 4.6.7.1.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE RD-433 DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

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4.6.7.1.1 RD-433 connections. Each RD-433 directly connected to an external power source and units deriving power from the RD-433 shall be placed on an insulating surface. All safety ground conductors between the RD-433 and units deriving power from the RD-433 shall be intact. The safety ground conductor between the RD-433 under test and the source power shall be opened during the test. The RD-433 shall be connected as shown on FIGURE 1 if it is connected to a single-phase source power.

4.6.7.1.2 Measurement. Leakage current shall be measured on the RD-433 in its normal operating configuration. RD-433 controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured for every combination of switch positions available on FIGURE 1. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.7 Special test routine. In order to simulate actual operation of the RD-433 during testing, a special test routine shall be used where specified herein. The special test routine shall be a computer program for the Data Processing Set, AN/UYK-20, which shall be used in conjunction with a video display terminal (VDT). The special test routine shall have the functions specified in a through c:

a. Write. Writing data into the RD-433, a sequence of data words is transferred from the computer to the RD-433. These sequences shall alternate between all zeroes, all ones, alternating zeroes and ones, and a pseudo-random sequence of ones and zeroes. The number of words to be transferred shall increase sequentially from one to 4096.

b. Read. For each write operation the data written into the RD-433 shall be read and verified. Should an error be detected, an indication that an error occurred shall be displayed on the VDT. Also displayed shall be the current word count, the sector address, what the data word actually is, and what it should be. The number of errors shall be accumulative and shall be displayed. The sequence of write and read shall be repeated continuously.

c. Printer option. The special test routine shall be capable of operating with a printer such that error conditions can be recorded over a long period of time.

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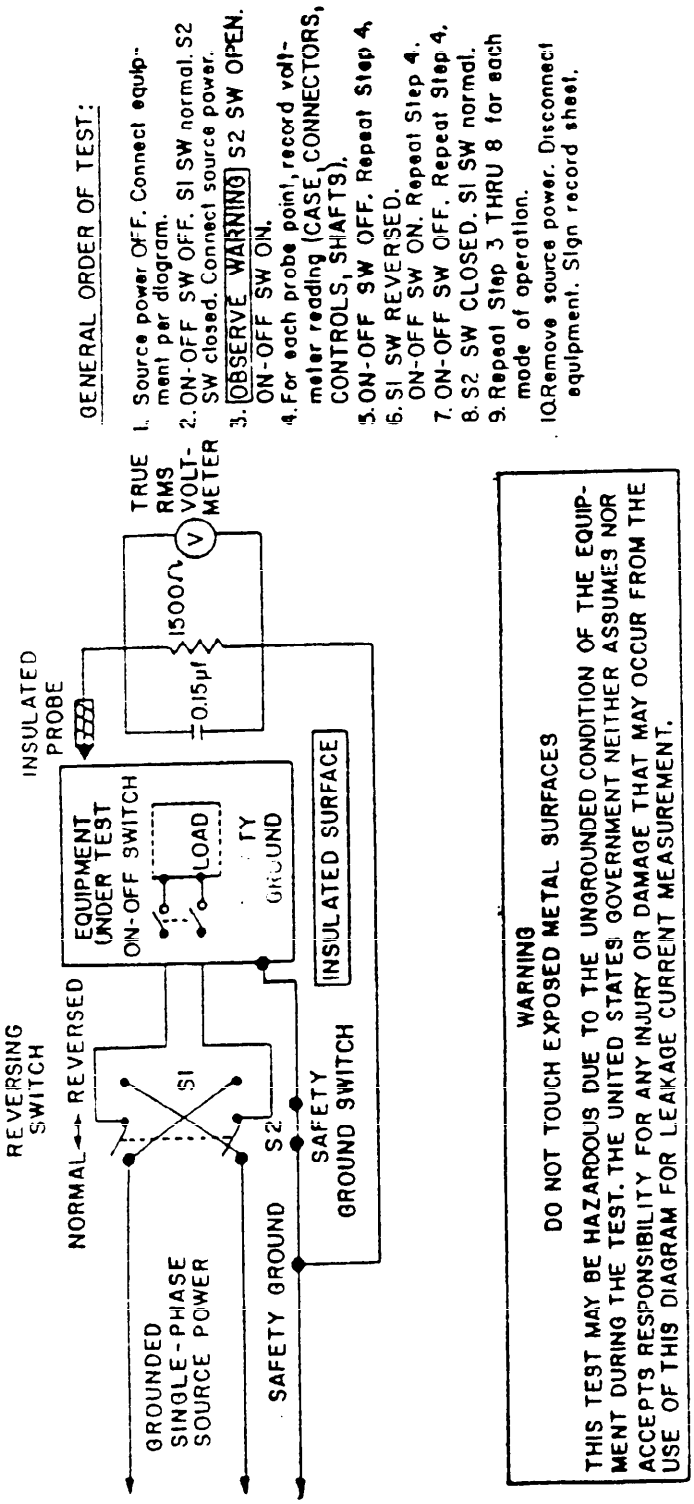


FIGURE 1. Single-phase test diagram for leakage current measurement.

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4.7.1 Government-furnished property. Equipment to support the testing specified in 4.7 will be Government-furnished property.

4.8 Reliability qualification tests. The RD-433 shall be tested to determine the conformance with 3.10. The reliability design qualification tests shall be in accordance with MIL-STD-781, Task 102, for Category 3A equipment.

4.9 Maintainability demonstration. The contractor shall perform a maintainability demonstration using diagnostics provided for this purpose to determine conformance to the quantitative maintainability requirements of 3.11. The demonstrations shall be performed by a technician with experience equivalent to that of an electronic technician third class (ET3).

4.9.1 Demonstration condition. The RD-433 maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by the replacement of subassemblies (modules, PCBs), chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts. The contractor shall determine 50 candidate faults in accordance with Appendix A of MIL-STD-471 for the demonstration. The procuring activity or its authorized representative shall use the candidate faults as a guide to determine a sample of 25 faults for the demonstration.

4.9.2 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the demonstration.

4.9.3 Technical documentation. Technical documentation shall be limited to the technical manual.

4.9.4 Acceptance criteria. Successfully located faults shall conform to the requirements of 3.11. Not more than two faults shall fail to be located.

4.9.5 Rejection. Failure to conform to the acceptance criteria or any of the maintainability requirements shall be cause for rejection of the demonstration. In the event of rejection, the contractor shall, at no additional cost to the Government, correct all deficiencies and repeat the demonstration.

4.10 Workmanship. Workmanship shall be as specified in 4.10.1 through 4.10.2.2.

4.10.1 General workmanship. The RD-433 shall be examined for conformance to the requirements of 3.12.1.

4.10.2 Workmanship screen. The workmanship screen shall be performed as specified in 4.10.2.1 and 4.10.2.2.

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4.10.2.1 Random vibration screen. Before conducting temperature cycling, vibration shall be performed on all RD-433s. The RD-433s shall be mounted in their normal operational position (without external shock isolators) and the vibration stress applied vertical to this position for an accumulated time of 10 minutes. The vibration shall be in accordance with FIGURE 2. Input vibration levels shall be measured at all mounting points of the RD-433 under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz, and a 100 Hz bandwidth above 500 Hz. The instantaneous acceleration peaks shall be limited to three times the rms acceleration level. The test RD-433 shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the functional characteristics. All failures occurring during the test shall require corrective action before the test is resumed.

4.10.2.2 Temperature cycling. Each RD-433 shall be subjected to 10 cycles of the temperature curve shown on FIGURE 3. The temperature rate of change shall be not less than 5°C per minute. RD-433 power shall be turned on and off at the indicated times. The RD-433 shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the RD-433 comes within 10°C of the chamber temperature. When failures occur, the RD-433 shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.11 Inspection of packaging. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

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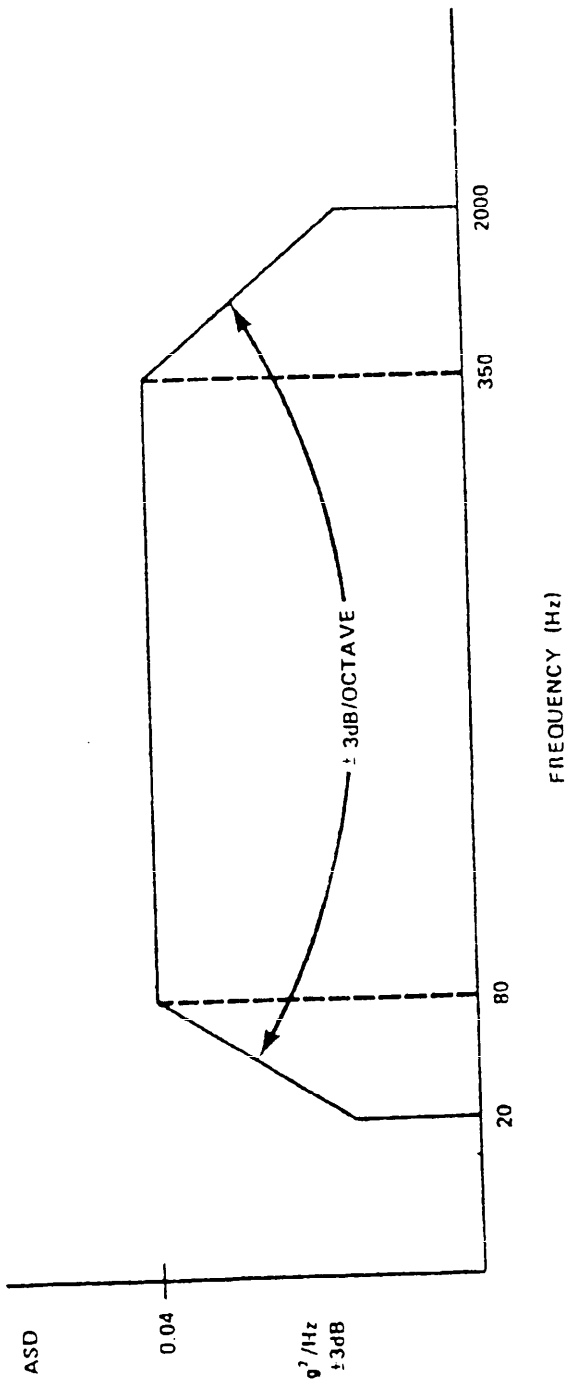


FIGURE 2. Random vibration curve.

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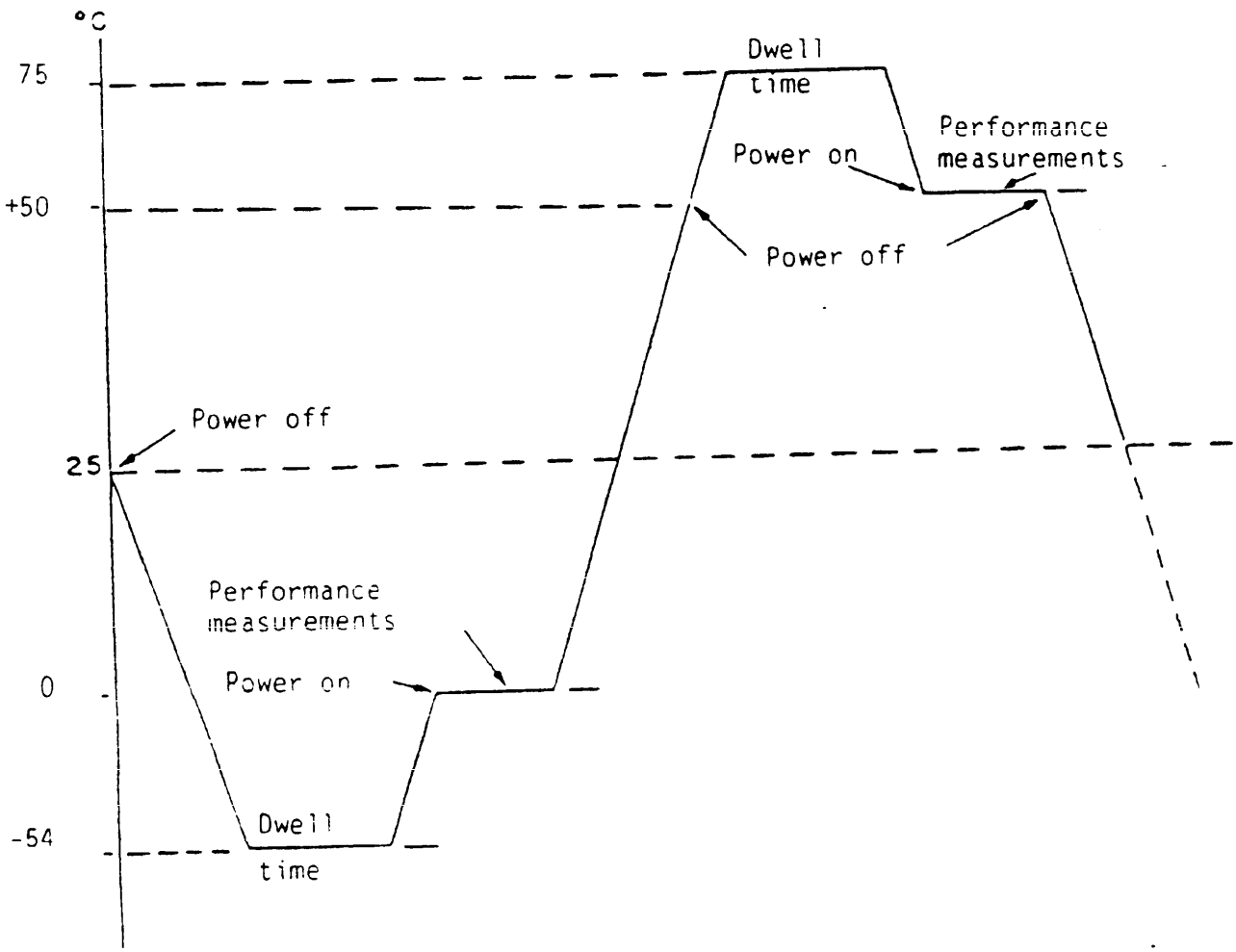


FIGURE 3. One cycle of temperature curve (not to scale).

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6. NOTES

6.1 Intended use. The RD-433, covered by this specification, is intended for use in the Tactical Information Exchange Subscriber Subsystem of the U.S. Navy Fleet Satellite Communication System. Other applications may include other AN/UYK-20-based systems used by the U.S. Navy.

6.2 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.4
c. When reinspected production control and environmental sample units may not be accepted (see 4.5.4)
d. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definition of a term used in this specification is given in 6.4.1.

6.4.1 Failure. Any malfunction or parameter deviation that prevents the RD-433 from performing within the operational requirements set forth in this specification.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 4.7.1.

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6.6 Subject term (key word) listing.

AN/UYK-20, Data Processing Set
Data Processing Set, AN/UYK-20
Intended use: In the Tactical Information Exchange Subscriber Subsystem of the U.S. Navy Fleet
Satellite Communication System
RD-433( )/SSH, Recorder-Reproducer, Digital
Recorder-Reproducer, Digital, RD-433( )/SSH

Preparing acitivity:
NAVY - EC

(Project No. 7025-N019(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-29530(EC)	Recorder-Reproducer, Digital RD-433( )/SSH

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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